

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ISLAMABAD

Annual Examination 2026
Biology — Class IX | Paper: SSC-I
Syllabus: Chapter 4 — Cell Cycle

Total Marks: 65

Time: 2h 30m

Version: 1

General Instructions:

1. Write your Roll Number in the space provided on the answer sheet. Do NOT write your name.
2. Attempt ALL questions from Section A. In Sections B and C, attempt the required number choosing either the main question OR its alternative (OR).
3. All parts of a question must be answered in sequence and at one place.
4. Extra attempt of any question or any part thereof will NOT be considered.

SECTION A — Multiple Choice Questions (12 × 1 = 12 Marks)

#	Question	A	B	C	D	Answer
1	The cell cycle consists of two main phases. These are:	☐ G1 and G2	☐ S phase and M phase	☐ Interphase and mitotic phase	☐ Karyokinesis and cytokinesis	
2	DNA replication takes place during which phase of interphase?	☐ G1 phase	☐ S phase	☐ G2 phase	☐ G0 phase	
3	During cell division, spindle fibres attach to a chromosome at the:	☐ Telomere	☐ Chromatid	☐ Centromere	☐ Nucleolus	
4	The longest phase of mitosis is:	☐ Metaphase	☐ Anaphase	☐ Telophase	☐ Prophase	
5	During metaphase of mitosis, chromosomes are aligned at the:	☐ Poles of the cell	☐ Equator of spindles	☐ Nuclear membrane	☐ Centromere region	
6	The spindle apparatus of plant cells differs from that of animal cells in not having:	☐ Microtubules	☐ Spindle fibres	☐ Centrioles	☐ Centromere	
7	A bivalent in meiosis consists of:	☐ Two chromatids, one centromere	☐ Four chromatids, two centromeres	☐ Two chromatids, two centromeres	☐ Four chromatids, four centromeres	
8	Crossing over occurs between non-sister chromatids during:	☐ Metaphase I	☐ Anaphase I	☐ Prophase I	☐ Interphase II	
9	Meiosis produces daughter cells that are:	☐ 2 diploid cells	☐ 4 diploid cells	☐ 2 haploid cells	☐ 4 haploid cells	
10	What causes the chromosome number to reduce to half when a cell divides by meiosis?	☐ DNA replication in interphase I	☐ Separation of homologous chromosomes in meiosis I	☐ Separation of sister chromatids in meiosis I	☐ Crossing over in prophase I	
11	The chromosomal number of fruit fly is 8. The gametes of fruit fly contain:	☐ 2 chromosomes	☐ 4 chromosomes	☐ 8 chromosomes	☐ 16 chromosomes	
12	The ability of a sea star to regenerate a lost arm depends on which type of cell division?	☐ Meiosis I	☐ Meiosis II	☐ Mitosis	☐ Binary fission	

SECTION B — Short Questions (11 × 3 = 33 Marks)

Attempt 11 questions. Choose either the main question OR its alternative.

Part	Question	Marks	OR	Alternative Question	Marks
2(i)	Define the cell cycle. Name its two main phases and state what happens in each.	3	OR	Describe the G0 phase. Which cells enter G0 permanently and which enter it temporarily? Give one example of each.	3
2(ii)	Describe the three sub-phases of interphase (G1, S, G2) and state the key event of each.	3	OR	What is the spindle apparatus? Describe how it is formed in animal cells and how plant cells differ in this process.	3
2(iii)	Describe the events of prophase in mitosis. What happens to the nucleolus and nuclear membrane?	3	OR	Describe the events of metaphase and anaphase in mitosis. What is the role of spindle fibres in anaphase?	3
2(iv)	What is telophase in mitosis? Describe the events that take place and what reforms at this stage.	3	OR	What is cytokinesis? Compare how it occurs in animal cells and plant cells, naming the structures involved.	3
2(v)	State four significances of mitosis in living organisms. Give one biological example for any one significance.	3	OR	What is meant by regeneration? Name three organisms capable of regeneration and the body part each can regenerate.	3
2(vi)	What is cancer? Distinguish between a benign tumour and a malignant tumour. What is metastasis?	3	OR	Define meiosis. State how it differs from mitosis in terms of number of divisions and daughter cells produced.	3
2(vii)	What is synapsis? What is a bivalent? Describe crossing over and define the term chiasma.	3	OR	Describe the events of prophase I in meiosis. Why is prophase I the longest phase of meiosis?	3
2(viii)	Describe metaphase I and anaphase I of meiosis. How does anaphase I differ from anaphase of mitosis?	3	OR	Describe telophase I and cytokinesis I in meiosis. What type of cells are produced after cytokinesis I?	3
2(ix)	Why is there no further DNA replication in interphase II? Describe what happens during meiosis II.	3	OR	Describe anaphase II and telophase II. How many cells are produced at the end of meiosis II and what is their ploidy?	3
2(x)	Explain how meiosis maintains the chromosome number in sexually reproducing organisms. Use human chromosomes (46) as your example.	3	OR	What are genetic variations? Explain two events of meiosis that are responsible for producing genetic variations.	3
2(xi)	Give any three differences between mitosis and meiosis from the textbook comparison table.	3	OR	Explain cloning and tissue culture. How has mitosis made these processes possible? Give one example of a plant used.	3

SECTION C — Long Questions (4 × 5 = 20 Marks)

Part	Question	Marks	OR	Alternative Question	Marks
3(i)	Describe the cell cycle in detail. Explain the four sub-phases of interphase (G1, S, G2, G0) including what occurs in each. What is the mitotic (M) phase?	5	OR	Describe the spindle apparatus. Explain its formation in animal and plant cells, its role in distributing genetic material, and what happens to it at the end of cell division.	5
3(ii)	Describe karyokinesis in mitosis. Explain all four stages (prophase, metaphase, anaphase, telophase) with the key events at each stage. Include the role of the centromere.	5	OR	Describe cytokinesis in mitosis for both animal and plant cells. Name all the structures formed (cleavage furrow, phragmoplast, middle lamella). State the significance of mitosis under four headings.	5
3(iii)	Describe meiosis I in detail. Explain interphase I, then all four stages of karyokinesis I (prophase I, metaphase I, anaphase I, telophase I), including synapsis and crossing over. What results after cytokinesis I?	5	OR	Describe meiosis II in detail. Explain interphase II, then all four stages of meiosis II (prophase II, metaphase II, anaphase II, telophase II), including cytokinesis. State the final result in terms of number and ploidy of cells.	5
3(iv)	Compare and contrast mitosis and meiosis. Give at least six differences between the two processes using any relevant criteria (cell type, number of divisions, daughter cells, chromosome number, crossing over, genetic similarity).	5	OR	Explain the significance of meiosis under two headings: (a) Maintenance of chromosome number in sexual reproduction, and (b) Genetic variations. Use the human example (46 chromosomes) and explain the role of crossing over in generating variation.	5

Invigilator's Signature: _____

Candidate's Signature: _____

Chapters Covered: Chapter 4 — Cell Cycle (Sections 4.1–4.5)

— End of Question Paper —